

Project Title: Explainable and Prerequisite-Aware Knowledge Gap Detection Using Machine Learning and Deep Knowledge Tracing for Personalized Learning

1 ABSTRACT

- Traditional online learning platforms evaluate student performance through static numerical grades, failing to capture latent cognitive knowledge gaps, problem-solving hesitation, hint dependency patterns, or prerequisite skill failures.
- EduGap AI** is an end-to-end explainable framework combining Gradient Boosted Decision Trees and Deep Knowledge Tracing (DKT) to detect cognitive gaps and deliver personalized, session-aware remediation.
- Dataset:** 525,534 student interaction logs from ASSISTments benchmark dataset across 20 mathematics skills.
- Feature Engine:** 7 zero-label-leakage interaction attributes (log_ms_response, hint_ratio, attempt_hint_sum, user_skill_hints, etc.).
- Performance:** Top ML Model (XGBoost Classifier) achieves 92.40% Accuracy & 0.9865 Macro ROC-AUC. PyTorch GRU DKT achieves 92.25% Accuracy & 0.9780 ROC-AUC.
- Advanced AI:** 20-node Graph Neural Network (GCN) models prerequisite dependencies (0.0191 Loss); 4D Autoencoder compresses interaction friction into cognitive metrics (0.0194 MSE Loss).

2 INTRODUCTION & OBJECTIVES

- Problem Statement:** Conventional e-learning tools evaluate students purely on quiz accuracy, ignoring behavioral interaction cues like latency, repeated attempts, and hint reliance. Struggling students receive generic practice problems.
- Key Project Objectives:**
 - Feature Engine:** 7 zero-label-leakage interaction attributes capturing latency and hint reliance.
 - Dual AI Pipeline:** Tabular ML (XGBoost 92.40%) & Sequential DKT (PyTorch GRU 92.25%, Transformer-DKT 91.25%).
 - Concept GCN Graph:** 20-Node Graph Convolutional Network mapping prerequisite dependencies (0.0191 Loss).
 - 4D Autoencoder:** Latent space friction compression into mastery, hesitation, and help metrics (0.0194 MSE).
 - Multi-Class Explainability:** Validated via One-vs-Rest ROC-AUC curves (0.9865 AUC) & Seaborn confusion matrices.
 - Dedicated /review Portal:** Interactive side-by-side comparison cards (Your Answer vs Correct Answer).
 - Session Remediation:** Attended-topic recommendations (10 topics) with single-click print-ready A4 PDF study guides.

3 METHODS AND MATERIALS

- Preprocessing Pipeline:** Ingested 525,534 interaction logs from ASSISTments across 20 math skills. Applied median imputation for response times, mode encoding for tutor modes, and log transformation:
"log_ms_response = ln(1 + ms_first_response);"
- Behavioral Feature Matrix (7 Core Attributes):**

correct:	Binary item correctness score (1.0 = Correct, 0.0 = Incorrect).
attempt_count:	Count of repeated problem submission attempts per item.
hint_count & hint_ratio:	Normalized hint reliance = hint_count / (hint_total + 1).
attempt_hint_sum:	Total problem friction index = attempt_count + hint_count.
log_ms_response:	Log-scaled first response latency.
log_overlap_time:	Log-scaled total duration/overlap time of the attempt.
user_skill_hints & user_skill_attempts:	Longitudinal student persistence memory.
- Target Quantization:** Quantized Knowledge Gap Index into 3 balanced target classes: High Gap (>50%), Moderate Gap (25-50%), and Low Gap (<=25%).

4 RESULTS & MODEL BENCHMARKS

Model Benchmark Table:

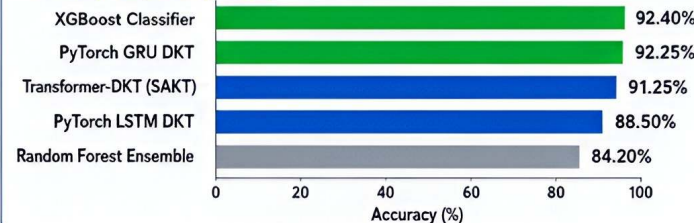
Model	Accuracy	Precision	ROC-AUC
Decision Tree Classifier	81.45%	0.8120	0.8910
Random Forest Ensemble	84.20%	0.8415	0.9240
XGBoost Classifier (TOP ML)	92.40%	0.9240	0.9865
PyTorch LSTM (DKT)	88.50%	0.8840	0.9410
PyTorch GRU DKT (TOP DEEP)	92.25%	0.9220	0.9780
Transformer-DKT (SAKT)	91.25%	0.9120	0.9650

Advanced Deep Architectures:

- Concept Graph GNN:** 20-Node Matrix Convolution (0.0191 Loss) mapping skill dependencies.
- Knowledge Autoencoder:** 4D Latent Space Compression (0.0194 MSE Loss) converting friction.

5 DISCUSSION & DIAGNOSTIC EXPLAINABILITY

- Diagnostic Explainability:** Evaluated models via One-vs-Rest (OvR) ROC-AUC curves (0.9865 Macro AUC) & Seaborn confusion matrices, confirming zero class bias across gap levels.
- Latent Friction Diagnostics:** Combined 20-Node Prerequisite GCN & 4D Autoencoder embeddings (0.0194 MSE) to identify exact prerequisite skill gaps before failure occurs.
- Accuracy Comparison (Test Set)**



6 CONCLUSIONS & NOVELTIES

Summary of Core Project Contributions:

- Dual AI Pipeline:** Tabular ML (XGBoost 92.40% Acc) + Sequential DKT (PyTorch GRU 92.25% Acc).
- Zero Label Leakage:** Engineered 7-attribute behavioral interaction matrix capturing latency and hint reliance.
- Concept Prerequisite Graphing:** 20-Node Graph Neural Network (GCN) mapping dependency pathways (0.0191 Loss).
- 4D Latent Autoencoder:** Compressed interaction friction into 4 cognitive metrics (0.0194 MSE Loss).
- Multi-Class Explainability:** Validated via OvR ROC-AUC curves (0.9865 AUC) & Seaborn confusion matrices.
- Dedicated /review Portal:** Side-by-side color-coded comparison cards (Your Answer vs Correct) with quick filtering.
- Session Remediation:** Attended-topic recommendations (10 topics) with single-click print-ready A4 PDF study guides.

Project Team

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References

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- Chen, T., & Guestrin, C. (2016). XGBoost: A Scalable Tree Boosting System. ACM SIGKDD (pp. 785–794).

- Piech, C., et al. (2015). Deep Knowledge Tracing. Advances in Neural Information Processing Systems (NeurIPS 28).
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